

# Adaptive agents in a coupled queueing model: phase separation, theory of mind, and the loss of universality

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Queueing models of human dynamics predict heavy-tailed interevent times, and extending them to two agents sharing a task that requires simultaneous execution yields a family of exponents  $\alpha = 1 + 1/\max_j(L_j - 1)$  set by the queue lengths [Oliveira and Vazquez, *Physica A* **388**, 187 (2009)]. In those models the priority assigned to the shared task is drawn from a fixed distribution. Here we let the agents *choose* it. Each agent assigns the interacting task a priority equal to its value discounted by the estimated risk that the partner will not participate, and estimates that risk from observation. The resulting system is a stag hunt with a saddle-node bifurcation: a coupled phase and an absorbing solitary phase, separated by a critical line  $R_c(L)$  that we obtain in closed form, valid only for  $L-1 \leq 1/c$ . We report five results. (i) Agents that additionally model *the partner's model of themselves* shift the critical line substantially, and a single such agent suffices to hold a non-modelling partner in the coupled phase. (ii) When the interaction opportunity is rare, coupling survives only if the partner model is held across silent intervals rather than decaying; a model maintained without input is what makes rare interaction viable. (iii) The scale-free interevent statistics of the non-adaptive model survive adaptation only inside a narrow window near the bifurcation, and only when agents retain a nonzero rate of uninformed exploration, without which the solitary phase is strictly absorbing. (iv) Inside that window the measured exponent is controlled by the fraction of time spent coupled and not by  $L$ : adaptation destroys the universality classes, and phase occupancy replaces  $\alpha$  as the natural order parameter. (v) At fixed parameters the outcome is bimodal but is not determined by early noise; collapse into the solitary phase is irreversible without intervention, and the partner-modelling capability that prevents collapse with certainty reverses it in only about a third of cases unless the agent is also willing to persist unilaterally for intervals far longer than a myopic evaluation sanctions, in which case recovery becomes complete. Allowing the agent to choose its own commitment length shows this policy to be optimal rather than imposed, and fixes its price: recovery requires persisting for  $0.74\tau_{\text{mem}}$  independently of how unfavourable the current estimate is, and an agent adopts it only if its horizon exceeds  $1.5\tau_{\text{mem}}$ . (vi) On a network, where each agent holds one interacting task per neighbour and attention is scarce, the solitary phase percolates as the cost of an unreciprocated offer grows, and high-degree nodes lose their relations first. Comparing a triadic-closure network with its degree-preserving randomization at matched coupled-edge fraction, local structure does not change *which* relations survive but does change how their failure is arranged: the coupled phase fragments into clusters under triadic closure while remaining connected without it. The transition is tractable analytically: attention divides as  $1/(k + a)$ , which fixes a critical degree  $k_c$  beyond which no coupled state exists, and percolation follows the Molloy–Reed criterion with the second moment truncated at  $k_c$ . Because the truncation acts at the top of the degree distribution, scale-free contact structures are maximally vulnerable.

## I. INTRODUCTION

Queueing theory has become a standard framework for the timing of human activity [1–3]. Representing an individual's to-do list as a priority queue with highest-priority-first execution produces power-law interevent time distributions  $P(\tau) \sim \tau^{-\alpha}$ , with  $\alpha = 1$  and  $\alpha = 3/2$  identified as universality classes and corroborated by electronic and postal correspondence data [1, 3, 4].

Single-queue models describe an isolated individual. Many activities, however, require two or more people at once, and interaction is known from the theory of critical phenomena to be decisive for universality [6]. A minimal two-agent extension [5] gives each agent one *interacting* task  $I$ , executable only when both agents select it in the same step, and one aggregate *non-interacting* task  $O$  standing for  $L_j - 1$  private tasks. That model predicts

a countable family of exponents

$$\alpha = 1 + \frac{1}{\max_j(L_j - 1)}, \quad (1)$$

interpolating between 2 and 1: the more distracted agent sets the exponent.

In that treatment the priority of the interacting task is drawn independently from a fixed density. This is the natural assumption if the shared activity is exogenous, but it removes the feature that makes joint tasks distinctive: an agent who is repeatedly stood up should stop offering. Here we make the interacting-task priority a decision variable. The agents estimate each other's participation rate from observation and set their priority accordingly. This changes the object from a stochastic process into a coupled inference problem and, as we show, changes its phenomenology substantially.

## II. MODEL

### A. Substrate

We retain the structure of Ref. [5]. Agent  $j \in \{A, B\}$  holds one interacting task  $I$  and one aggregate task  $O$  representing  $a_j \equiv L_j - 1$  private tasks. The private-task priority is the maximum of  $a_j$  independent uniform variates,

$$f_{O_j}(x) = a_j x^{a_j-1}, \quad x \in [0, 1], \quad (2)$$

redrawn whenever  $O$  is executed. Each agent selects the higher-priority task.  $I$  executes only if both agents select it.

### B. Priority as a decision

Let  $x_{I_j}$  be the priority agent  $j$  assigns to the interacting task. Since  $O$  is drawn afresh each step, the probability that  $j$  selects  $I$  is

$$p_j = \Pr(x_{O_j} < x_{I_j}) = x_{I_j}^{a_j}. \quad (3)$$

The agent controls a propensity, not a decision: with many private tasks,  $a_j$  large, even a high  $x_{I_j}$  yields a small participation probability.

Payoffs per step: both agents select  $I$ , each receives  $R_I$ ; an agent selects  $I$  while its partner does not, it receives  $x_{O_j} - c$ , the step being wasted and  $c > 0$  the cost of an unreciprocated offer; otherwise it receives  $x_{O_j}$ , the realised value of its private task. The cost  $c$  is what makes a model of the partner necessary; at  $c = 0$  the optimal policy is  $x_{I_j} = 1$  regardless of the partner.

Writing  $\hat{p}$  for the agent's estimate of its partner's participation probability, selecting  $I$  is preferable whenever  $\hat{p}R_I + (1 - \hat{p})(x_{O_j} - c) > x_{O_j}$ , giving the optimal priority

$$x_{I_j}^* = \left[ R_I - c \frac{1 - \hat{p}}{\hat{p}} \right]_0^1, \quad (4)$$

where  $[\cdot]_0^1$  denotes clipping to  $[0, 1]$ : *the value of the interaction discounted by the risk that the partner will not participate*. Estimates are maintained by an exponential moving average with retention  $\lambda = e^{-1/\tau_{\text{mem}}}$  over the partner's observed selections.

### C. Levels of modelling

We distinguish three policies by what each agent represents.

**Level 0:** No partner model:  $x_I = R_I$ , clipped.

**Level 1:** Partner model:  $\hat{p}$  tracked, Eq. (4) applied.

**Level 2:** Self model: the agent additionally tracks  $\hat{s}$ , its estimate of *the partner's estimate of itself*, and evaluates a finite-horizon rollout of the coupled dynamics in which its own participation raises  $\hat{s}$ , which raises the partner's priority through Eq. (4), which raises the partner's participation. It commits to  $x_I = 1$  when that rollout dominates the myopic choice.

Level 2 is the minimal policy that requires an agent to represent itself as an object in another agent's model.

### D. Exploration

Each agent additionally participates with probability  $\epsilon$  per step irrespective of belief. This is not a numerical convenience. At  $\epsilon = 0$  the state  $p = 0$  is strictly absorbing: an agent that stops participating never observes its partner participating and therefore never revises. Escape requires both agents to explore in the same step, at rate  $\epsilon^2$ .

## III. FIXED POINTS

For a symmetric pair the participation probability obeys  $p \mapsto \mathcal{F}(p)$  with

$$\mathcal{F}(p) = \left( \left[ R_I - c \frac{1-p}{p} \right]_0^1 \right)^a. \quad (5)$$

Setting  $\mathcal{F}(p) = p$  and requiring tangency gives the saddle-node in closed form,

$$p^* = (ac)^{a/(a+1)}, \quad R_c = (p^*)^{1/a} + \frac{c}{p^*} - c. \quad (6)$$

For  $c = 0.35$  this gives  $R_c = 0.8332$  at  $p^* = 0.5916$  for  $L = 2$ , and  $R_c = 0.9818$  at  $p^* = 0.7885$  for  $L = 3$ .

Equation (6) is meaningful only while  $p^* \leq 1$ , that is

$$a \leq 1/c. \quad (7)$$

Above this bound the tangency lies outside the accessible range and there is no interior bifurcation: the coupled phase is created discontinuously at the clipping boundary  $x_I = 1$ . At  $c = 0.35$  this restricts the continuous transition to  $L \leq 3.86$ . Figure 1 shows the return map and the critical line.

Below  $R_c$  only the solitary fixed point  $p = 0$  exists. Above it there are two stable phases, coupled and solitary, separated by an unstable point whose position rises with  $a$ : more private tasks enlarge the basin of solitude. The structure is that of a stag hunt, with the solitary phase risk-dominant and absorbing at  $\epsilon = 0$ .

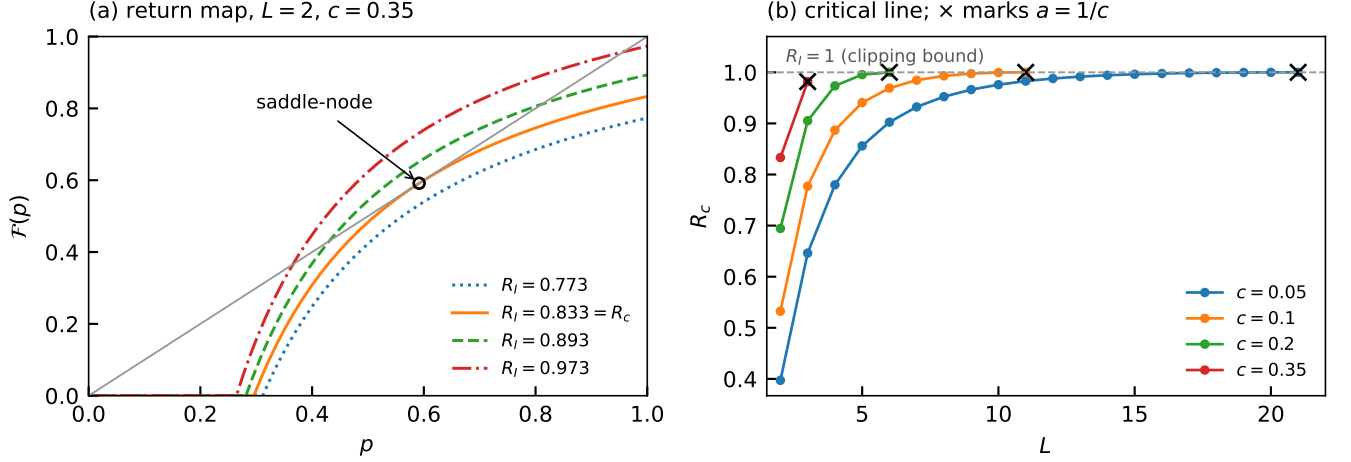


FIG. 1. (a) Return map  $\mathcal{F}(p)$  for  $L = 2$ ,  $c = 0.35$  at several  $R_I$ , with the diagonal in grey. The two nontrivial fixed points are born in a saddle-node at  $R_c = 0.8332$ ,  $p^* = 0.5916$  (circle). (b) Critical line  $R_c(L)$  from Eq. (6) for several costs  $c$ ; crosses mark the validity boundary  $a = 1/c$ , Eq. (7), beyond which no interior bifurcation exists.

#### IV. RESULTS

Unless stated otherwise, simulations use  $\tau_{\text{mem}} = 200$ , symmetric agents, and  $\geq 10^5$  steps after a 20% burn-in. We quantify the phase by the *occupancy*  $\phi$ , the fraction of time with  $p > p^*/2$ .

##### A. Validation

Reproducing Ref. [5] with i.i.d. priorities recovers Eq. (1). Asymmetric pairs sit on the prediction:  $(L_A, L_B) = (2, 3)$  gives  $\alpha = 1.500$  and  $(2, 5)$  gives  $\alpha = 1.250$ , both stable across four decades of the tail. Symmetric pairs approach from below —  $L = 2$  gives 1.82 at the 90th percentile, drifting to 1.94 deeper in the tail — consistent with the logarithmic correction expected when  $a_A = a_B$ . We also confirm that the mean interevent time diverges with the observation window as  $T^{2-\alpha}$ , so that for  $\alpha \leq 2$  the interaction possesses no characteristic timescale.

##### B. Modelling the partner's model shifts the critical line

Level-2 agents remain coupled well below the level-1 threshold. At  $L = 3$  a level-1 pair collapses to the solitary phase for every  $R_I \leq 0.99$ , whereas a level-2 pair remains coupled down to  $R_I = 0.90$ . The effect is markedly asymmetric in a way that inverts the  $\max_j$  rule of Eq. (1): at  $L = 3$  and  $R_I \in \{0.94, 0.96, 0.98\}$ , pairs of level-1 agents collapse while a level-2/level-1 pair remains coupled (Table I). In the non-adaptive model the more distracted agent sets the exponent; in the adaptive model the more sophisticated agent sets the phase.

##### C. Rare interaction requires a persistent partner model

We next decouple the rate of *opportunity* from willingness by allowing an interaction to be possible only with probability  $\nu$  per step, and by making the partner's action unobservable except when an interaction occurs. The mean gap between opportunities,  $1/\nu$ , then sets the rate at which information about the partner arrives.

If unobserved steps decay the estimate — silence treated as refusal — the coupled phase is lost whenever the estimate comes to be dominated by accumulated silence rather than by the last observed encounter. At  $1/\nu = 100$ , pairs with  $\tau_{\text{mem}} \in \{5, 50\}$  remain coupled while  $\tau_{\text{mem}} \in \{500, 5000\}$  collapse.

The decisive variable, however, is not the ratio of timescales but whether the estimate is held at all. If unobserved steps leave  $\hat{p}$  unchanged, coupling survives at *every* opportunity rate and every memory constant tested, with the realised interaction rate tracking  $\nu$  to three digits (Table II). A partner model maintained through intervals containing no information is what makes rare interaction viable.

TABLE I. Asymmetric pairing at  $L = 3$ ,  $\tau_{\text{mem}} = 50$ . A single level-2 agent holds the pair in the coupled phase.

| $R_I$ | levels ( $A, B$ ) | phase    | occupancy |
|-------|-------------------|----------|-----------|
| 0.94  | (1,1)             | solitary | 0.000     |
| 0.94  | (2,1)             | coupled  | 0.942     |
| 0.96  | (1,1)             | solitary | 0.000     |
| 0.96  | (2,1)             | coupled  | 0.961     |
| 0.98  | (1,1)             | solitary | 0.000     |
| 0.98  | (2,1)             | coupled  | 0.980     |

TABLE II. Occupancy at  $L = 3$ ,  $R_I = 0.99$ , level 2, with the partner’s action unobservable between interactions. Left: silence decays the estimate. Right: the estimate is held.

| $1/\nu$ | decaying                 |       | held                     |       |
|---------|--------------------------|-------|--------------------------|-------|
|         | $\tau_{\text{mem}} = 50$ | 5000  | $\tau_{\text{mem}} = 50$ | 5000  |
| 10      | 0.843                    | 0.000 | 1.000                    | 1.000 |
| 100     | 0.756                    | 0.000 | 1.000                    | 1.000 |
| 333     | 0.755                    | 0.000 | 1.000                    | 1.000 |

#### D. Adaptation destroys the universality classes

Away from  $R_c$  the adaptive system is not scale-free: it polarises to  $\tau = 1$  in the coupled phase and  $\tau \rightarrow \infty$  in the solitary one, with apparent tail indices in the range 4–40. Heavy tails reappear only in a narrow window above  $R_c$ , where the two fixed points nearly coincide and belief noise drives intermittent switching. Inside that window the interevent coefficient of variation reaches 8–50 and the tail is favoured over an exponential by log-likelihood differences of  $10^3$  to  $10^5$ .

The window exists only for  $\epsilon > 0$ . At  $\epsilon = 0$  the solitary phase is absorbing, the pair falls once and never returns, and no value of  $R_I$  produces a heavy tail.

Crucially, the exponent measured in the window is *not* given by Eq. (1). Because the transition is sharp,  $\phi$  cannot be tuned through  $R_I$ ; varying only the noise realisation at a fixed operating point, however, spreads  $\phi$  over a wide range, and  $\alpha$  tracks it (Table III). The correlation between  $\phi$  and  $\alpha$  is +0.70 at  $L = 2$  and +0.76 at  $L = 6$ ; the two  $\alpha(\phi)$  curves nearly coincide, while Eq. (1) would require 2.00 and 1.20 respectively.

We conclude that  $\alpha$  is not an order parameter of the adaptive system. The occupancy  $\phi$  is the natural replacement: bounded, directly measured, and monotone in  $R_I$ ,  $L$  and modelling level, with the scale-free interevent statistics appearing as a symptom of intermediate  $\phi$  rather than as its definition.

TABLE III. Tail index versus occupancy at fixed operating points, varying only the random seed. Rows at  $\phi = 1$  have CV  $\simeq 0.76$  and are not heavy-tailed; their fitted indices are reported for completeness only.

|                                | $L = 2, c = 0.35$ | $L = 6, c = 0.10$ |
|--------------------------------|-------------------|-------------------|
| Eq. (1)                        | 2.00              | 1.20              |
| corr( $\phi, \alpha$ )         | +0.70             | +0.76             |
| $\alpha$ at $\phi \simeq 0.22$ | 1.79              | —                 |
| $\alpha$ at $\phi \simeq 0.68$ | 4.59              | 4.32              |
| $\alpha$ at $\phi = 1.00$      | 4.62–4.65         | 4.37–4.42         |

#### E. Bimodality, irreversibility, and the asymmetry of prevention

At fixed parameters ( $L = 2$ ,  $c = 0.35$ ,  $R_I = 0.8428$ ,  $\epsilon = 10^{-2}$ ) the outcome over 140 realisations is strongly bimodal: 45 end with  $\phi < 0.25$ , 93 with  $\phi > 0.75$ , and only 2 in between.

The outcome is nevertheless not fixed by early noise. Classifying the final phase from a trajectory prefix barely exceeds the majority-class baseline of 0.679: accuracies are 0.679, 0.693, 0.693, 0.714, 0.736 and 0.836 for prefixes of 1, 3, 10, 30, 60 and  $120 \times 10^3$  steps. The first  $10^4$  steps carry essentially no information. Collapse is a rare event available throughout the trajectory rather than a bifurcation resolved at the outset.

Collapse is, however, irreversible without intervention: none of the 45 collapsed realisations recovered when left alone. Promoting agents to level 2 at the midpoint recovers 35.6% of them, whereas pairs of level-2 agents run from  $t = 0$  on the same seeds never collapse at all (Table V). Promoting both agents is worth nothing over promoting one: recovery is driven by a single agent participating against the advice of its own estimate, and the partner need only be able to respond.

#### F. The recovery cap is a limit of patience, not of reachability

The partial recovery in Table V might indicate states from which no policy in this family can return. It does not. As specified above, the level-2 rollout evaluates a single step of unreciprocated participation. We therefore introduce a commitment length  $k$ : the number of consecutive steps of unilateral participation the agent evaluates, and latches onto if it decides to proceed.

The decision is deterministic given beliefs, so the reachable region can be mapped directly. Table IV reports the smallest  $\hat{p}$  at which an agent still chooses to participate, taking  $\hat{s} = \hat{p}$ . At  $k = 1$  the agent participates only if it already believes its partner participates  $\gtrsim 46\%$  of the time — a condition a collapsed pair fails by construction. Increasing  $k$  alone is insufficient, and increasing the lookahead  $H$  alone changes nothing at all: recovery is flat at 48.5% for  $H \in \{10, \dots, 1000\}$  at  $k = 1$ . Only when a long commitment is combined with a discount horizon  $1/(1 - \gamma)$  long enough to see past it does the threshold collapse to zero, and then every collapsed realisation recovers.

The residual 48.5% is not policy-driven: it is the fraction of realisations in which exploration happens to lift  $\hat{p}$  above the participation threshold, at which point the agent latches. No state is therefore beyond recovery. What the cap measures is the reluctance of a short-horizon agent to persist without evidence. Note that the baselines in Tables V and IV (35.6% and 48.5%) come from runs of different length with different collapsed subsets and are not directly comparable.

TABLE IV. Smallest belief  $\hat{p}$  at which the agent still participates (left), and measured recovery of the collapsed realisations (right), versus commitment length  $k$  and discount  $\gamma$ . Recovery is 48.5% in every cell but one.

| $k$  | min $\hat{p}$   |                  | recovered       |                  |
|------|-----------------|------------------|-----------------|------------------|
|      | $\gamma = 0.97$ | $\gamma = 0.999$ | $\gamma = 0.97$ | $\gamma = 0.999$ |
| 1    | 0.463           | 0.458            | 48.5%           | 48.5%            |
| 10   | 0.458           | 0.453            | 48.5%           | 48.5%            |
| 100  | 0.428           | 0.313            | 48.5%           | 48.5%            |
| 1000 | 0.428           | $10^{-6}$        | 48.5%           | 100%             |

### G. Endogenous commitment: the price of recovery is a memory time

The commitment length above is imposed. We now let the agent evaluate its rollout at each candidate length and latch onto the argmax, so that the length becomes a property of its beliefs. The argmax is uninformative — it pins to the largest candidate, since once the partner participates further length is free — so we characterise the policy by the minimum patience that beats not persisting,

$$k_{\min}(\hat{p}) = \min\{k : \mathcal{R}(k) > \mathcal{R}(0)\}, \quad (8)$$

with  $\mathcal{R}(k)$  the rollout value under a  $k$ -step commitment.

Two regularities emerge, both in units of the memory constant  $\tau_{\text{mem}}$ . First,  $k_{\min}$  does not diverge as  $\hat{p} \rightarrow 0$ ; it saturates, and bisecting at  $\hat{p} = 10^{-6}$  gives  $k_{\min}/\tau_{\text{mem}} = 0.740, 0.740, 0.740, 0.738$  for  $\tau_{\text{mem}} = 50, 100, 200, 400$ . The price of recovery is therefore a fixed multiple of a memory time and is independent of how unfavourable the current estimate is. The related constant  $0.347 \tau_{\text{mem}}$  is where the partner begins to respond: its myopic priority turns positive at  $s > c/(R_I + c) = 0.293$ , reached at  $-\tau_{\text{mem}} \ln(1 - 0.293)$ . Persistence must continue about twice that long before the accumulated gain covers the commitment.

Second, whether the agent will pay that price is set by its horizon in the same units: the smallest  $H^* = 1/(1 - \gamma)$  at which an agent with  $\hat{p} = 10^{-6}$  still commits is  $H^* \simeq 1.5 \tau_{\text{mem}}$  (measured 1.50 at  $\tau_{\text{mem}} = 100, 200, 400$ ; the value 2.00 at  $\tau_{\text{mem}} = 25, 50$  is an upper bound set by the resolution of the  $H$  ladder).

These predict the simulation. At  $\tau_{\text{mem}} = 200$ , so  $H^* \simeq 300$ , an agent choosing its own commitment length recovers 48.5% of collapsed realisations at  $\gamma \leq 0.995$  ( $H \leq 200$ ) and 100% at  $\gamma = 0.999$  ( $H = 1000$ ), while the fixed  $k = 1$  control stays at 48.5% throughout. Judging recovery instead by interaction rate — an unbiased measure, since the occupancy  $\phi$  exceeds  $1/2$  whenever one agent is latched irrespective of its partner — moves the baseline to 72.7% but leaves the contrast unchanged, at 100% for the same single condition, with the mean interaction rate rising from 0.368 to 0.639 and the partner's

TABLE V. Intervention on the 45 realisations that collapsed as level-1 pairs, with promotion at the midpoint of the run.

| intervention                  | recovered | mean final $\phi$ |
|-------------------------------|-----------|-------------------|
| none (control)                | 0.0%      | 0.020             |
| agent $A \rightarrow$ level 2 | 35.6%     | 0.369             |
| both $\rightarrow$ level 2    | 35.6%     | 0.369             |
| level 2 from $t = 0$          | 100%      | 1.000             |

payoff from 0.642 to 0.734 against a solitary baseline of  $\langle x_O \rangle = 0.5$ . The recoveries are genuine couplings.

The agent that restores the interaction is not the one that gains most from it: on recovered realisations the committing agent earns 0.62–0.70 against its partner's 0.68–0.76, the difference being the cost  $c$  of the unreciprocated offers it absorbs on the way.

## V. NETWORKS: PERCOLATION OF THE SOLITARY PHASE

The two-agent model extends to a network by giving agent  $i$  one interacting task per neighbour alongside its solitary aggregate, with a single item selected per step. Attention then becomes a scarce resource: an agent with  $k$  relations leaves  $k - 1$  offers unanswered at every step, and each relation must compete with the others as well as with the private queue. An edge executes, as before, only when both endpoints select it.

Two features of the queueing substrate turn out to be essential here and are worth stating, since abandoning either produces an artefact. First, a task's priority must be *drawn* from a distribution the policy controls and held until execution, rather than being a deterministic function of  $\hat{p}$ . With a deterministic priority each agent locks onto its single best partner and every other relation starves, so the coupled subgraph is forced to be a perfect matching. Second, an *offer* must consume the attempt whether or not it is reciprocated. Redrawing only on execution freezes the choice: agent  $i$  offers to the same partner indefinitely, and if that partner's best is not  $i$  the pair deadlocks with no mechanism for change.

We take the growth rules from Refs. [10, 11]: triadic closure  $LS(\ell = 1)$ , the undirected local-search rule in which a newcomer links to a random node and to one of its neighbours, closing a triangle at every step; its degree-preserving randomization; and  $BA(m = 2)$  as a non-local control. The first is local and has a finite Ramsey number  $r_k = 81$ , so communities are certain beyond a few hundred nodes; the other two have none. Randomizing  $LS$  preserves the degree sequence exactly, so any difference between the two is attributable to local structure alone.

We use  $c$ , the cost of an unreciprocated offer, as the control parameter, and call an edge coupled when its interaction rate exceeds ten times the solitary baseline  $\epsilon^2$ . The order parameter  $S$  is the largest connected com-

TABLE VI. Coupled edge fraction and giant component  $S$  of the coupled subgraph, at  $R_I = 0.95$ ,  $n = 600$ ,  $\langle k \rangle = 4$ , averaged over three realisations.

| $c$   | $LS(\ell = 1)$ |              | randomized |              | $BA(m = 2)$ |       |
|-------|----------------|--------------|------------|--------------|-------------|-------|
|       | edges          | $S$          | edges      | $S$          | edges       | $S$   |
| 0.002 | 0.988          | 1.000        | 0.990      | 1.000        | 0.940       | 1.000 |
| 0.010 | 0.950          | 1.000        | 0.955      | 1.000        | 0.874       | 1.000 |
| 0.020 | 0.686          | <b>0.569</b> | 0.689      | <b>0.976</b> | 0.605       | 0.967 |
| 0.030 | 0.183          | 0.006        | 0.178      | 0.006        | 0.165       | 0.007 |
| 0.070 | 0.000          | 0.000        | 0.000      | 0.000        | 0.000       | 0.000 |

ponent of the coupled subgraph as a fraction of  $n$  (Table VI).

The solitary phase percolates between  $c = 0.02$  and  $c = 0.03$ , and the interesting behaviour is at the lower edge of that window. There the two networks have statistically identical coupled edge fractions, 0.686 against 0.689, and identical degree-resolved survival, yet entirely different connectivity: the triadic-closure network has  $S = 0.569$ , thirteen components of size three or more, and a second component of 99 nodes, while its randomization remains essentially intact at  $S = 0.976$  with a second component of 4. Local structure does not determine which relations survive; it determines how their failure is arranged. Under triadic closure the coupled phase fragments into separated clusters, whereas at identical degrees without local structure it thins while staying connected. The split between local and non-local rules that governs the emergence of communities in Ref. [10] thus reappears as a split in the way the solitary phase invades.

Survival falls monotonically with degree — the fraction of a node's relations that remain coupled at  $c = 0.020$  is 0.856, 0.704, 0.539 and 0.346 for degrees near 2, 4, 8 and 16 — and the two networks agree to within 0.01 at every degree. This follows directly from the attention constraint:  $\hat{p} \simeq P(\text{active})/k$ , so a relation is sustainable only while  $k \lesssim P(\text{active})(R_I + c)/c$ . High degree is a liability rather than a protection, because what an agent competes for is its partner's attention, and the coincidence of the two curves shows the effect to be independent of topology.

### A. Analytic threshold for arbitrary degree distribution

The transition admits a closed-form treatment. Since priorities are  $uX$  with  $u \sim U(0, 1)$  drawn per offer and the solitary task has  $F_O(t) = t^a$ , the probability that an agent of degree  $k$  selects one particular neighbour is

$$\int_0^1 u^{k-1} (uX)^a du = \frac{X^a}{k+a}, \quad (9)$$

so attention divides as  $1/(k+a)$ . Self-consistency  $X = \Psi(X^a/(k+a))$  with  $\Psi(p) = B - c/p$  and  $B = R_I + c$  gives  $X^{a+1} - BX^a + c(k+a) = 0$ , whose tangency at  $X^* = aB/(a+1)$  fixes a critical degree

$$k_c = \frac{a^a(R_I + c)^{a+1}}{c(a+1)^{a+1}} - a, \quad (10)$$

equal to  $(R_I + c)^2/4c - 1$  for  $a = 1$ . Beyond  $k_c$  no coupled state exists at any belief: attention is divided  $k$  ways,  $\hat{p} \sim 1/k$ , and it falls below the collapse threshold  $c/(R_I + c)$ .

An edge is live only if sustained from both ends, so the process is degree-targeted site percolation and the Molloy–Reed criterion applies with the second moment truncated at  $k_c$ ,

$$\sum_{k \leq k_c} k(k-1)p_k > \langle k \rangle. \quad (11)$$

Equation (11) uses the full degree, but a node whose relations have already failed divides its attention only among the survivors. Writing  $\phi$  for the probability that an edge is sustained from one end and  $q_k = kp_k/\langle k \rangle$ , the cavity equation closes on a locally tree-like graph as

$$\phi = \sum_k q_k \mathbb{P}[\text{Bin}(k-1, \phi) \leq k_c - 1], \quad (12)$$

with a giant component iff  $b = \sum_k q_k \mathbb{E}[(m-1)\mathbf{1}\{m \leq k_c\}]/\phi > 1$ ,  $m = 1 + \text{Bin}(k-1, \phi)$ . Setting  $\phi = 1$  recovers (11), which is therefore an upper bound. Exploration at rate  $\epsilon$  sets  $x = 1$  and hence wins the selection, consuming the attention slot with probability  $1 - (1 - \epsilon)^k$ ; including it replaces  $c(k+a)$  by  $c(k+a)(1 - \epsilon)^{-k}$  in (10).

Equation (10) reproduces the measured degree dependence: survival falls monotonically with  $k$ , and  $k_c = 10.8$  at  $c = 0.020$  sits where the measured survival halves. The threshold itself, however, is a mean-field upper bound. On the randomized network, where the tree-like assumption holds, Eq. (12) gives  $c_c \simeq 0.062$  against a measured  $c_c \simeq 0.025$  — long runs place the transition sharply between  $c = 0.020$  (coupled fraction 0.825,  $S = 0.997$ ) and  $c = 0.030$  (0.106,  $S = 0.007$ ). The discrepancy is the expected one: a mean-field treatment cannot capture fluctuation-driven escape into an absorbing state, and that escape is precisely the stag-hunt mechanism, the runaway in which a falling  $\hat{p}$  lowers  $X$ , which reduces the offers received, which lowers  $\hat{p}$  again. Relations that (10) declares viable are destroyed by it.

Two practical points follow. Convergence is slow and, near the transition, directional: over  $2 \times 10^4$  to  $3.2 \times 10^5$  steps the coupled fraction *rises* from 0.488 to 0.802 at  $c = 0.020$  while *falling* from 0.165 to 0.082 at  $c = 0.030$ , so a threshold read from short runs is biased toward the initial condition. And the truncation in (11) acts at the *top* of the degree distribution. For  $p_k \sim k^{-\gamma}$  with  $2 < \gamma < 3$  the second moment diverges and the network is robust to random removal, but the removal here is targeted at high degree — the one attack to which such

networks are most fragile. A scale-free contact structure is therefore maximally vulnerable to this form of collapse, for a mechanical rather than a topological reason: what an agent competes for is its partner's attention, so high degree is a liability.

We report the transition location but claim no exponent:  $n = 600$  with three realisations is too small, and the agents here are all level 1, the network analogue of the self-model being left for future work.

## VI. DISCUSSION

Making the shared-task priority a decision variable converts the two-agent queueing model into a coordination game with an absorbing failure mode, and the consequences are not confined to the transition region. The countable universality classes of Eq. (1) do not survive: heavy tails persist only in a narrow window near the saddle-node, and there the exponent is governed by phase occupancy rather than by queue length. The exponents reported for correspondence data should therefore be read as evidence that the underlying priority assignment is close to exogenous. Where participation is itself strategic, the same measurement is not expected to yield  $1 + 1/\max_j(L_j - 1)$ .

Two structural results appear robust. First, the capacity to represent a partner's representation of oneself enlarges the coupled phase, and a single such agent stabilises a pair. Second, in a system whose failure mode is absorbing, the same capacity is far more effective as prevention than as cure unless it is combined with a will-

ingness to persist unilaterally over intervals much longer than a myopic evaluation would sanction; recovery then becomes complete, but it requires acting against one's own current estimate.

Both results, together with the requirement that a partner model be sustained through intervals containing no observations, were reached in the course of examining a specific hypothesis: that experience is a property of systems whose preferred states can only be reached through another system they do not fully control. Under that reading the coupled phase is the object of interest, the solitary phase is its absence, and  $\phi$  measures the distance between them. We record the motivation for transparency; nothing in Secs. II–IV depends on it, and we do not claim the model bears on the question.

Several directions remain. The two constants 0.74 and 1.5 are expressed in units of  $\tau_{\text{mem}}$ , but the agent propagates its partner's estimate using its own memory constant, and all runs reported here use  $\tau_A = \tau_B$ ; whether the partner's true constant or the agent's assumption about it governs recovery requires asymmetric memory and a per-agent diagnostic. Whether deeper recursion continues to enlarge the coupled phase, or saturates at finite depth, would determine whether modelling depth is a useful graded quantity. Finally, the extension to  $N$  agents on a network is immediate in principle, and the question of whether the solitary phase percolates is well posed.

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